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Reducing Food Waste in Campus Dining: A Data-Driven Approach to Demand Prediction and Sustainability

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Abstract: Tracking density in universities is essential for planning services like food, transportation, and social activities on campus. However, food waste remains a critical challenge in campus dining operations, leading to significant environmental and economic consequences. Addressing this issue is crucial not only for minimizing environmental impact but also for achieving sustainable operational efficiency. Campus food services significantly influence students' university choices; thus, forecasting meal consumption and preferences enables effective planning. This study tackles food waste by analyzing daily campus data with machine learning, revealing strategic insights related to food variety and sustainability. The algorithms Linear Regression, Extra Tree Regressor, Lasso, Decision Tree Regressor, XGBoost Regressor, and Gradient Boosting Regressor were used to predict food preferences and daily meal counts. Among these, the Lasso algorithm demonstrated the highest accuracy with an R^2 metric value of 0.999, while the XGBRegressor also performed well with an R^2 metric value of 0.882. The results underline that factors such as meal variety, counts, revenue, campus mobility, and temperature effectively influence food preferences. By balancing production with demand, this model significantly reduced food waste to 28%. This achievement highlights the potential for machine learning models to enhance sustainable dining services and operational efficiency on university campuses.

Keywords: campus mobility; food prediction; food waste reduction; machine learning; meal variety

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1. Introduction

The mobility of student and staff numbers in universities and higher education institutions is an important indicator of campus density. This density requires strategic planning and resource allocation to effectively manage security, mobility, energy consumption, food needs, and other related issues in various areas of the campus [1]. Campus food services are of great importance within the university and serve as a significant factor in the preferences of prospective students [2]. University campus food services typically consist of different areas such as the central dining hall and cafeteria, which provide services to meet the food needs of students and staff with various options [3]. However, insufficient management of food services often leads to significant food waste, which not only increases operational costs but also contributes to environmental degradation, making this an urgent issue for sustainable campus management. Therefore, the quality and variety of food services directly affect the satisfaction and efficiency levels of campus life [4,5].

Artificial intelligence (AI) and machine learning (ML) enable more efficient and effective delivery of various services on university campuses [6]. AI and ML optimize the use of resources and increase student and staff satisfaction to ensure efficiency and sustainability

in the services provided [7]. Although studies such as those by Yang et al. (2022) and Lorenzoni et al. (2021) have explored aspects like clustering consumption patterns or analyzing transaction data, these approaches often fail to integrate environmental metrics and broader sustainability goals [8,9]. Within the scope of food services, one of the most important services provided at the university, it is observed that machine learning methods are effective in the management and planning of campus food services [10]. Despite this, existing research lacks a comprehensive framework that connects campus density, dining preferences, and sustainability outcomes, leaving critical gaps in understanding the systemic impact of food waste in universities.

ML has also been instrumental in improving sustainability efforts in food waste management by enabling better prediction models and inventory management [11]. In addition, the quality of food service in campus areas is an influential factor in satisfying students and staff [12], making it crucial that catering operations are carried out with this quality in mind [13,14]. ML has been utilized to optimize food quality prediction, enhance quality control processes for food products, and predict shelf life. Specifically, the Random Forest Optimizer method was applied to forecast food freshness in storage areas, achieving an impressive accuracy of 94.01% [15]. It is also applied in food identification processes such as food recognition, calorie measurement, and calorie calculations [16]. Particularly, ML optimizes the use of raw materials by increasing food production efficiency, analyzing market dynamics, enhancing consumer preferences, and improving inventory management by forecasting demand [7,17].

In the literature, there are studies examining various aspects of catering services on university campuses. For example, Yang et al. (2022) analyzed canteen consumption data using k-means clustering and large learning network architectures to identify students in need of financial aid. Lorenzoni et al. (2021) applied a multivariate clustering method to data from cashier transactions at a university in Pisa, demonstrating that students prefer processed food and proteins [8,9]. Demirel and Türk (2024) analyzed the demographic structure of Iğdır University students and their knowledge about organic products, revealing a significant relationship between organic products and physical health [18]. Furthermore, Rahman (2024) evaluated students' cafeteria satisfaction and its influencing factors at a public university in Bangladesh, employing machine learning Recursive Feature Elimination (RFE) methods [19].

Research on demand and forecasting models developed to analyze consumer preferences, food preferences, and meal variety highlights the importance of these aspects in reducing waste and enhancing sustainability [20]. Martin-Rios et al. (2020) also emphasized that sustainability-oriented innovations in food waste management technology are critical for reducing waste and aligning with circular economy principles in campus food services [9]. One of the previous studies addressing food waste reduction in campuses examined the impact of plate shape and size on minimizing waste. Richardson et al. (2021) used machine learning algorithms to predict meal demand and demonstrated that oval plates significantly reduced food waste. These findings highlight the importance of physical factors that complement machine learning-based meal demand predictions [20]. In another study, the effects of tray and tray-less dining systems on university students' food choices, consumption, and waste behaviors were analyzed, revealing significant implications for food management [21]. Aci and Yergök (2023) developed demand forecasting models for university dining halls, incorporating the calendar effect and turnstile passage data, and examined eighteen Machine Learning models categorized under five main techniques. Among these, the EDT Boosted model demonstrated the most effective prediction performance, achieving metrics such as MSE = 0.51, MAE = 0.50, and R = 0.96, highlighting its superiority in forecasting meal demand [22].

Finally, some studies explore the broader effects of university dining services on student satisfaction and well-being through surveys and analytical methods [23–25]. Simulation-based approaches, such as Kambli et al. (2020), revealed that customer waiting times can be reduced by 45% through optimized capacity allocation and queue management [3]. Additional research by Oliveria et al. (2024) assessed dietary preferences among Portuguese university students, highlighting vending machine consumption and lunchbox usage rates. Sakai et al. (2022) examined unhealthy eating habits and nutrient deficiencies in Indonesian canteen menus [26,27].

Research demonstrates that the efficiency, student satisfaction, and sustainability of university dining services have been extensively investigated through various studies. These studies utilize diverse datasets, including meal types, number of meals, calorie counts, and food preferences. While satisfaction surveys are commonly employed to evaluate dining experiences, the development of automated systems that integrate real-time food preferences and consumption data offers a more effective approach. Although such surveys provide valuable insights, machine learning models frequently address isolated issues and establish limited relationships with a narrow set of variables, thereby diminishing their overall impact on comprehensive campus management strategies.

Sustainability in universities goes beyond reducing food waste; it aligns with the global movement towards a circular economy and resource efficiency. Addressing food waste within campus systems not only mitigates environmental harm but also serves as a model for broader societal sustainability practices. This study, therefore, bridges these gaps by combining machine learning models with campus-specific data to create actionable insights, contributing to reduced waste and enhanced operational efficiency.

In this study, cafeteria turnover data, dining hall food types, number of meals, campus density, and daily temperature values were comprehensively analyzed. Machine learning algorithms were applied to predict outcomes, such as the most preferred food types and the number of meals required for dining services. Furthermore, the daily and hourly flow of people entering the campus and food consumption patterns were examined to assess their influence on students' food preferences. These analyses identified the most favored food items, predicted meal quantities, and highlighted preferences in consumption areas using high-performance models. The findings contribute to reduced food waste and enhanced sustainability in campus dining services.

2. Materials and Methods

2.1. Dataset

In the research, the dataset was created by selecting the year 2024, spring semester, when the education and training process is the busiest in Suleyman Demirel University Campus. The data consists of daily records of various activities and parameters within the campus. Campus entrance information, university dining hall entrance information (staff and student), and university commercial cafeteria turnover information were obtained from the university. In addition to these data, the database was enriched by obtaining publicly published weather information, food types in the dining hall, and calorie information. This database, which contains detailed information such as campus mobility (215,504 people), number of meals (55,006 pieces), daily activities, etc., was analyzed to determine the effects of campus density, food preferences, cafeteria performance, and weather.

The dataset is divided into several main categories: temporal, spatial, meteorological, food, user statistics, and financial information. Temporal data includes detailed timestamps such as date, day, month, year, hour, and minute. Spatial information includes door movements reflecting campus entry and exit movements. Meteorological data provides information on weather conditions that may affect campus activities by recording the average

daily maximum and minimum temperatures. Furthermore, daily menu information forms the food data category, while user statistics segment the population into students, academic staff, and administrative staff. Finally, financial data is represented by the turnover figures of the campus cafeteria. On the data obtained, missing data were edited, categorical data were transformed, and data were scaled.

Table 1 provides an overview of all the variables used in the analysis, along with clear descriptions of their roles and significance in the context of the study. This detailed breakdown ensures transparency and highlights the data-driven, systematic approach of the study. By including a comprehensive description of the dataset, this work establishes a solid foundation for understanding and addressing food waste on campus.

Table 1. Descriptive features of the dataset.

Feature Name	Description
GATE	Gate entry point for data collection.
MOVEMENT	Indicator of movement type (e.g., entry, exit).
ID	Unique identifier for records.
DATE	Date and time of the recorded data.
MONTH NAME	Name of the month in the recorded data.
WEEKDAYS	Day of the week in the recorded data.
AVERAGE TEMPERATURE	Average temperature on the recorded day.
MAIN LUNCH 1	Primary dish served for lunch.
MAIN LUNCH 2	Secondary dish served for lunch.
MAX TEMPERATURE	Maximum temperature on the recorded day.
MIN TEMPERATURE	Minimum temperature on the recorded day.
ADDITIONAL PRODUCTS	Additional products available during lunch.
SOUP	Type of soup served for lunch.
NOON STUDENT	Number of students during noon hours.
NOON ADMINISTRATIVE	Number of administrative staff during noon hours.
TOTAL DAILY PERSONS	Total number of people on the recorded day.
CALORIES (KCAL)	Calories of the meal in kilocalories.
STAFF LUNCH	Number of staff members having lunch.
STAFF ENTRY	Staff members' entry count.
CANTEEN REVENUE	Revenue generated by the canteen.
TOTAL DAILY MOVEMENT PERSON	Total movement of persons on the recorded day.
MENU	Details of the menu served.
INVERSE CANTEEN REVENUE	Inverse value of canteen revenue for some analysis.

2.2. Feature Extraction

The process of creating numerical features from raw, unprocessed data without compromising dataset integrity is known as feature extraction. Feature selection is an important step in the development of machine learning models. The number of input features can range from two to hundreds of features, many of which may be insignificant or have a lower correlation with the target variables. Feature selection methods improve the performance of models on high-dimensional datasets by reducing training time, improving generalization, and enhancing interpretability [28]. Model parameters were optimized with feature engineering to enhance model performance. The menus of the days of the week were obtained by web scraping, and the minimum and maximum temperature values for those days were added with the same method. This data was used as a feature for a machine learning model or analysis. In addition, the need for categorization of the menus was identified in the initial analysis, and the success of the model was increased by adding new features in the process.

During the feature extraction phase, the missing values in the data set were identified in the missing data editing stage, and the data were completed. In the data transformation

phase, the data should be converted into numerical formats that can be used by regression algorithms. In this process, categorical data were converted into numerical data with the one-hot encoding method. The unit matrix provides many mathematical and structural advantages in data transformations and algorithms. The superiority of the data over each other was removed. This was achieved by associating the data with the unit matrix. When columns are in string format, converting with one-hot encoding often results in vectors consisting of a large number of zeros. In such cases, a "sparse matrix" was used to store the data more efficiently. The sparse matrix reduces memory usage by storing only positions that are "1".

2.3. Preprocessing Using StandardScaler

Standardization is a data preprocessing step that allows data to scale within a certain range and speeds up algorithm calculations. In particular, this process transforms the features in the dataset to have a mean of zero and a standard deviation of one. To do this, the mean is subtracted from each dataset and the result is divided by the standard deviation. Commonly used scalers for transforming data sets include the robust scaler, min-max scaler, normalizer, and standard scaler. The following formula (Equation (1)) can be used to make adjustments: σ is the standard deviation, μ is the mean, and W is the initial value [29].

$$Y = \frac{W - \mu}{\sigma} \quad (1)$$

Data scaling, transformation and feature reduction create a standardized data set, reducing prediction errors. The accuracy of models is affected by processes such as feature selection, data cleaning, and scaling. Estimating the number of meals per day is achieved through standardization in data preparation, which can be achieved using tools such as StandardScaler. By normalizing the data within a certain range, consistent measurements across variables are ensured. This process was applied to prevent overfitting and under fitting.

2.4. Machine Learning Model and Regression

Regression algorithms aim to predict the value of the dependent variable using one or more independent variables [30]. With this data set, regression algorithms can make various predictions [31]. In this study, a total of 5 machine learning prediction algorithms were trained considering the characteristics of the dataset, and their success metrics and test graphs were interpreted and analyzed. The selection of these algorithms was based on the structural characteristics of the dataset (numerical and categorical variables), the type of relationship with the dependent variable (linear and nonlinear relationships), and the flexibility and accuracy of the models.

The following features were included as input parameters: 'average temperature', 'max temperature', 'min temperature', 'noon student', 'noon administrative', 'calories (kcal)', 'staff lunch', 'staff entry', 'canteen revenue', and 'menu'. These variables were selected for their predictive relevance to the target variable. The primary target parameter was 'total daily dining persons', representing the total number of individuals interacting with the system daily. This parameter is essential for understanding campus movement and dining requirements.

The dataset was divided into training (80%) and test (20%) subsets to effectively evaluate the models' learning and prediction performance on unseen data. This partitioning ensures that the models generalize their learning and maintain accuracy on data not included in the training process. Feature importance and transformations played a significant role in the analysis. For instance, "average temperature" was identified as a key predictor in models such as Lasso regression, where its coefficient highlighted a strong influence on

the target variable. To ensure optimal contribution to prediction accuracy, the "average temperature" feature was mathematically transformed by multiplying it by a factor of 10. In this context, coefficients represent the extent to which a unit change in an independent variable affects the target outcome, emphasizing why certain features are more critical in predictive modeling.

2.4.1. Gradient Boosting Regressor

Gradient Boosting Regressor is an ensemble learning method that builds a strong predictive model by sequentially training weak learners (usually decision trees). Each step minimizes cumulative errors, ensuring high accuracy [32]. This algorithm was selected due to its ability to handle both numerical (e.g., temperature, total movement) and categorical (e.g., gate type, meal type) variables effectively. Tree-based models like Gradient Boosting excel in modeling complex nonlinear relationships, such as predicting meal counts. Its robust performance and low error rates make it ideal for this dataset.

2.4.2. XGBoost Regressor

XGBoost Regressor is a high-performance gradient boosting algorithm known for its speed, scalability, and detailed hyperparameter control. It is effective on large datasets and ensures high accuracy [33]. The algorithm was chosen for its ability to handle both numerical (e.g., temperature) and categorical (e.g., gate type) variables. Its scalability and capacity to model nonlinear patterns make it ideal for predicting meal counts with high efficiency and accuracy.

2.4.3. Linear Regression

Linear Regression is a basic regression model widely used in statistics and machine learning and is used to model the linear relationship between the dependent variable and independent variables; basically, it allows predicting the target variable with a linear combination of features (predictor variables) given as input, and this model is used to understand the relationship between the variables in the data set, make predictions, and analyze the effects of the variables; its performance varies depending on the linear relationship of the features in the data set and the suitability of the model [34]. This algorithm was selected for its simplicity and ability to analyze linear trends in numerical variables (e.g., temperature, total movement). It provides interpretable results and efficiently captures primary relationships in the dataset.

2.4.4. Linear Regression Lasso

Lasso (Least Absolute Shrinkage and Selection Operator) regression is a technique for feature selection and regularization in linear regression models. This method reduces some regression coefficients to zero to reduce the complexity of the model and minimize the risk of overfitting. Thus, only the most important variables are retained in the model, which increases its simplicity and interpretability [35]. Lasso regression is a powerful technique to avoid feature selection and overfitting, especially in high-dimensional data sets. When the regularization parameter is chosen appropriately, the Lasso regression model yields simpler, more interpretable, and highly generalizable results [36–38]. Lasso Regression was selected due to the dataset's mix of numerical and categorical variables, many of which might be irrelevant or have weak correlations with the target variable. Its feature selection capability makes it ideal for reducing the dataset's dimensionality, focusing on the most impactful variables. This not only improves model generalization but also prevents overfitting, ensuring more robust predictions for variables like total daily movement or meal counts.

2.4.5. Decision Tree Regressor

The Decision Tree (DT) is an efficient algorithm for solving classification and regression problems. Its basic principle is to break down complex problems into simpler sub-problems, making the solution easier and more understandable. DT is based on hierarchically organized rules from root to leaf, and these rules provide the predictive power of the model [28]. DT offers the flexibility to work with both categorical and numerical data, and provides a better understanding of the results by using if-else rules in the decision-making process. However, there is a risk of overfitting if the model is subjected to excessive branching; therefore, the generalizability of the model can be increased through different processes such as pruning or cross-validation [39]. DT was selected for its ability to handle both numerical (e.g., "Average Temperature") and categorical (e.g., "Weekdays") variables. It creates interpretable rules, such as predicting "Total Daily Persons" based on meal types and days. Its flexibility with mixed data types and ability to model complex patterns make it suitable for this dataset. To prevent overfitting, pruning and cross-validation can enhance its generalizability.

2.4.6. Extra Tree Regressor

Extra Tree Regressor is part of the Extremely Randomized Trees (Extra Trees) algorithm, one of the ensemble methods, and is used in regression problems. This model prevents overlearning and increases generalization ability by applying randomization when building decision trees. Extra Trees is an extension of the random forest algorithm, splitting nodes with randomized features and values, and works on the entire training dataset, while random forest uses a bootstrap replica [40]. Extra Tree Regressor was preferred for its effectiveness in handling numerical (e.g., "average temperature") and categorical (e.g., "weekdays") variables in the dataset. Its randomized splits reduce overfitting and enhance generalization, making it ideal for robust predictions such as "total daily persons.

2.5. Validation Methods

Cross-validation methods were applied to evaluate the performance and generalization ability of the models. This technique splits the dataset into training and testing subsets and evaluates model performance on each subset individually to ensure reliable results. Different fold counts were used to compare the models' performance based on metrics such as R^2, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) [22]. To reduce data imbalance and the risk of overfitting, the data was randomly selected, and different training and testing combinations were created in each validation cycle. The validation process contributed to a detailed analysis of each algorithm's performance and improved the reliability of the proposed methods. This approach aimed to enhance prediction accuracy and establish a solid foundation for sustainable dining services.

3. Results

3.1. Analysis of Relationships in Campus Dataset Results

As the central tendency and spread of the variables in the dataset, the hourly movement of people entering the campus is given in Figure 1. In this context, the hourly density analysis on campus will contribute to the planning to be made in areas such as food services, security needs, development of internet infrastructure, cleaning and maintenance services, technological infrastructure, transportation and traffic management, academic services, and social and cultural activities, and will enable optimum service provision and effective management on campus.

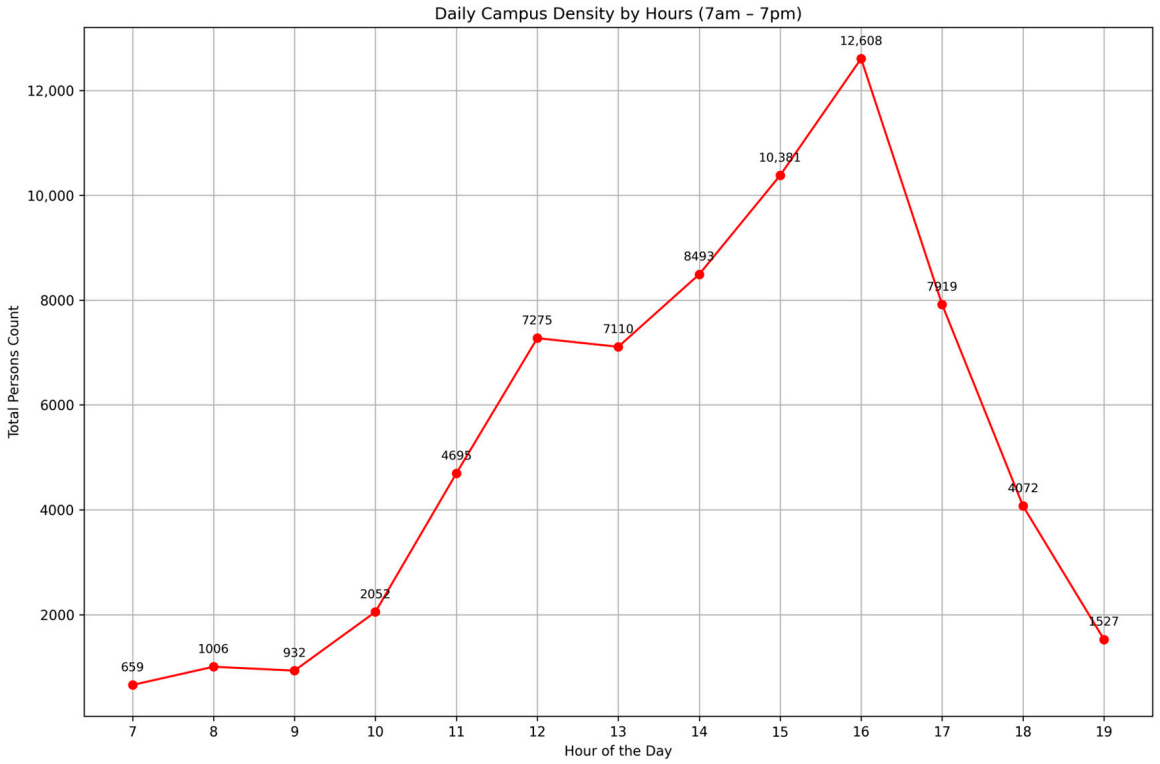


Figure 1. Daily campus density by hours.

The relationship between the total number of people on campus and dining hall usage according to the daily menu is shown in Figure 2. In the analysis, which was sorted according to the most preferred menu in the dining hall, it was revealed that the use of the dining hall did not increase linearly on days when the total daily movement persons count was higher. It was determined that the food menus primarily affect the dining hall preference.

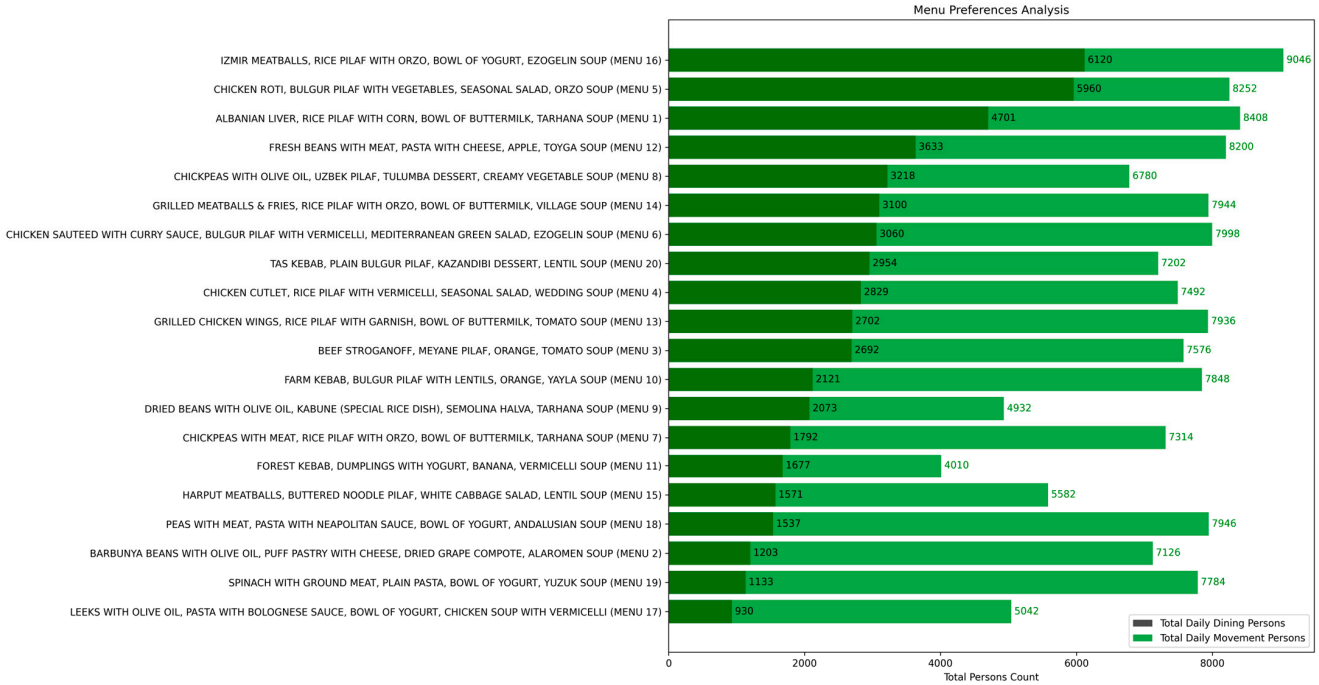


Figure 2. Menu preferences analysis based on campus density.

The relationship analysis between dining hall usage and cafeteria usage is shown in Figure 3. It has been observed that an increase in the total daily dining count is associated with a linear decrease in the average cafeteria revenue.

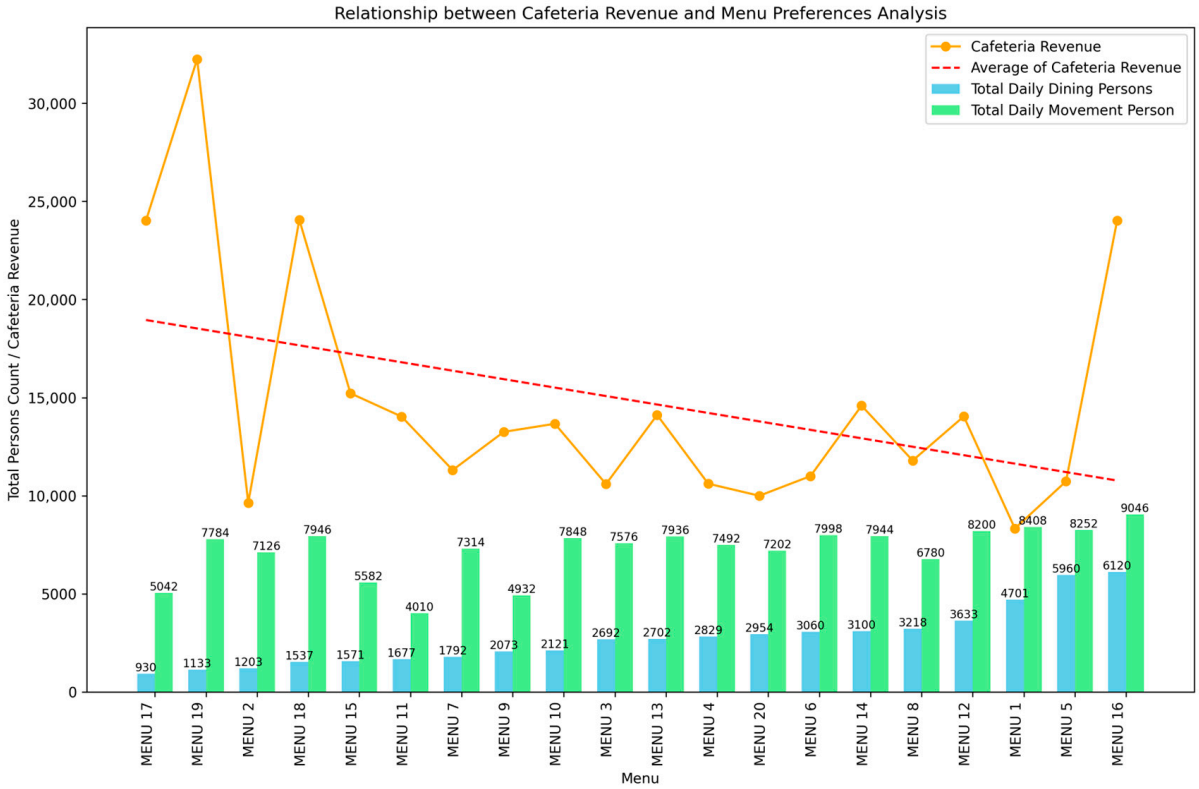


Figure 3. Relationship between cafeteria revenue and dining hall usage.

In the analysis, the expected linear increase in dining hall usage was not observed on days with high daily mobility. This suggests that the main factor determining dining hall preferences is the menus offered rather than the intensity of movement. Menus stood out as a factor that significantly influenced students' choices, and therefore, menus need to be categorized and analyzed in more depth.

The meals served in the university dining hall are prepared daily in four types: two main courses, a soup, and one additional product. In order to determine the preference rates within the scope of the research, one menu data including meal types were first divided into four categories: main course one, main course two, soup, and additional product. The preference frequencies of the meals within each category were analyzed to determine which meals were most preferred. The preference rates of 19 different types of food in the categories of main course one and main course two, which are preferred depending on daily consumption, are given in Figures 4 and 5.

There are 14 and 12 different options in the soup, and supplement categories offered in the food menus. Figures 6 and 7 show the graphs of soup preference and supplement preference rates.

The correlation heatmap, which visually represents the correlations between the variables in the dataset, is illustrated in Figure 8. Positive Correlations: if there is a red cell, this indicates that these two variables have a strong positive relationship. Negative Correlations: if there is a dark blue cell in between, this indicates that these two variables have a strong negative relationship. When the correlation matrix between average temperature, total daily movement person, cafeteria revenue, and total daily dining count in the campus dataset is examined, it shows that there is a very weak and negative relationship of

−0.05 between average temperature and daily dining count. However, temperature has a strong relationship with total daily movement person.

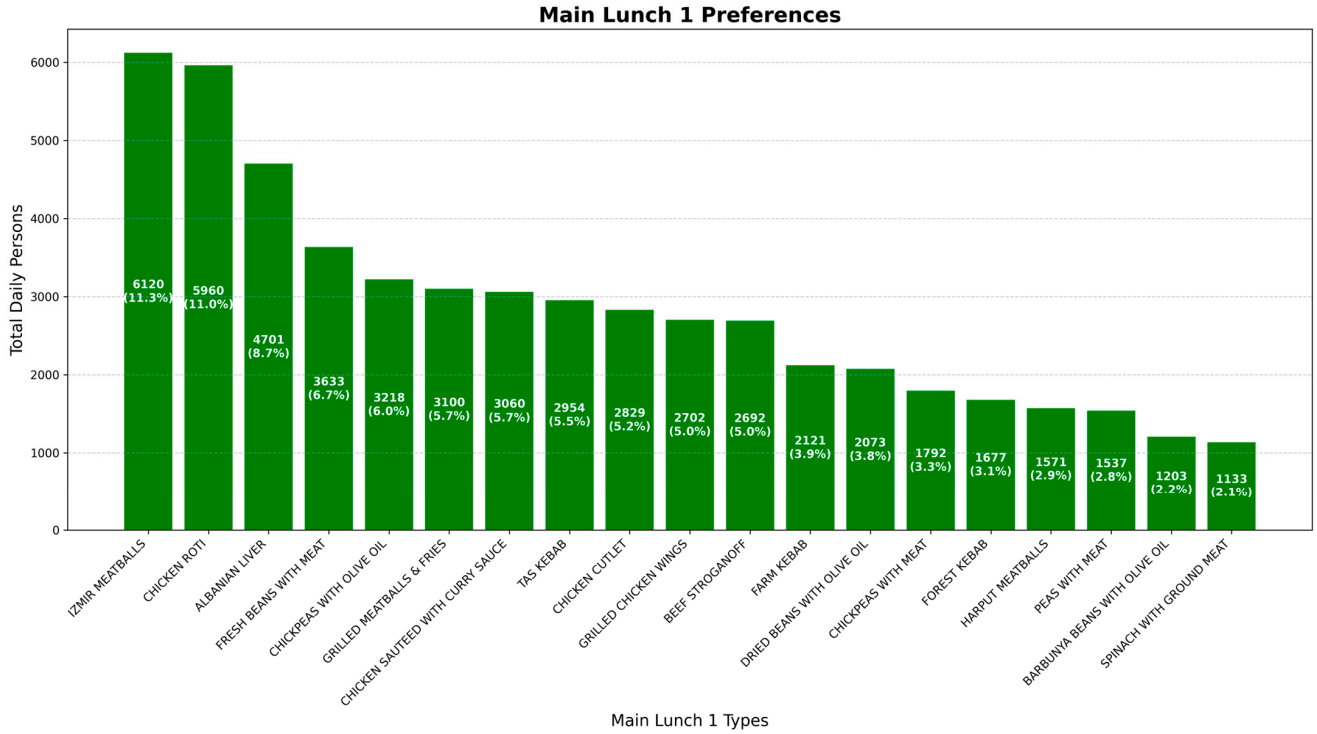


Figure 4. Graph of preference rates for main lunch 1.

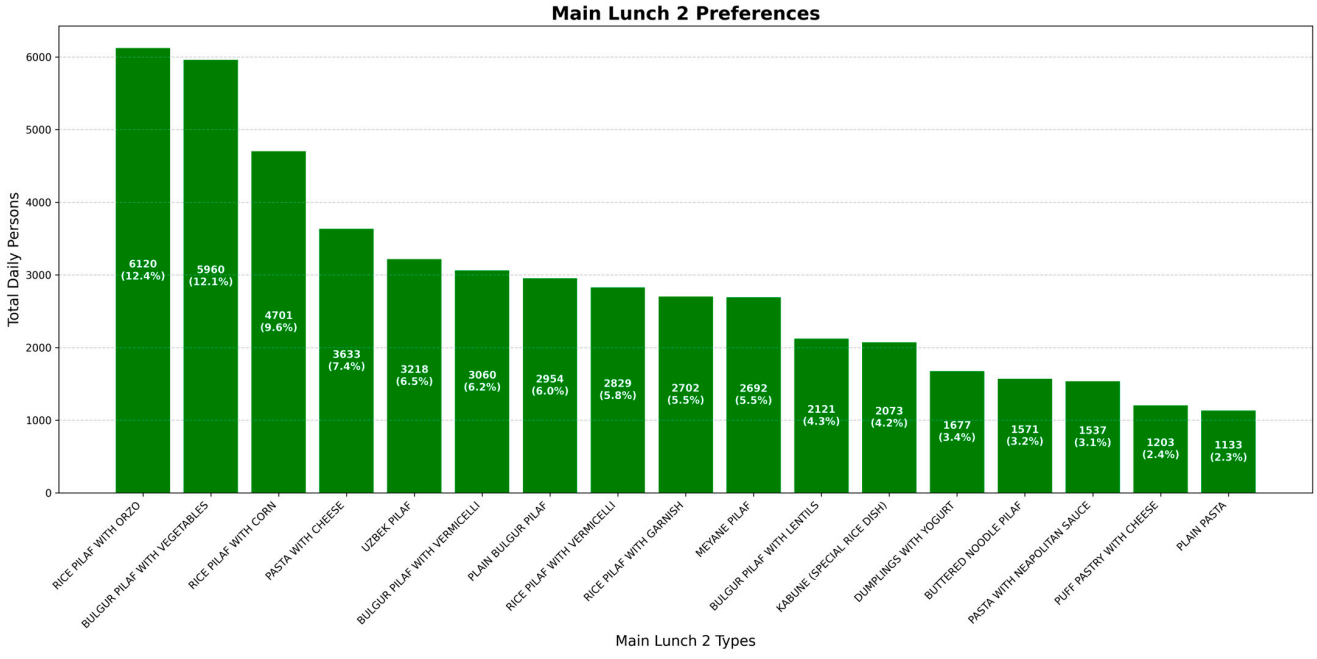


Figure 5. Graph of preference rates for main lunch 2.

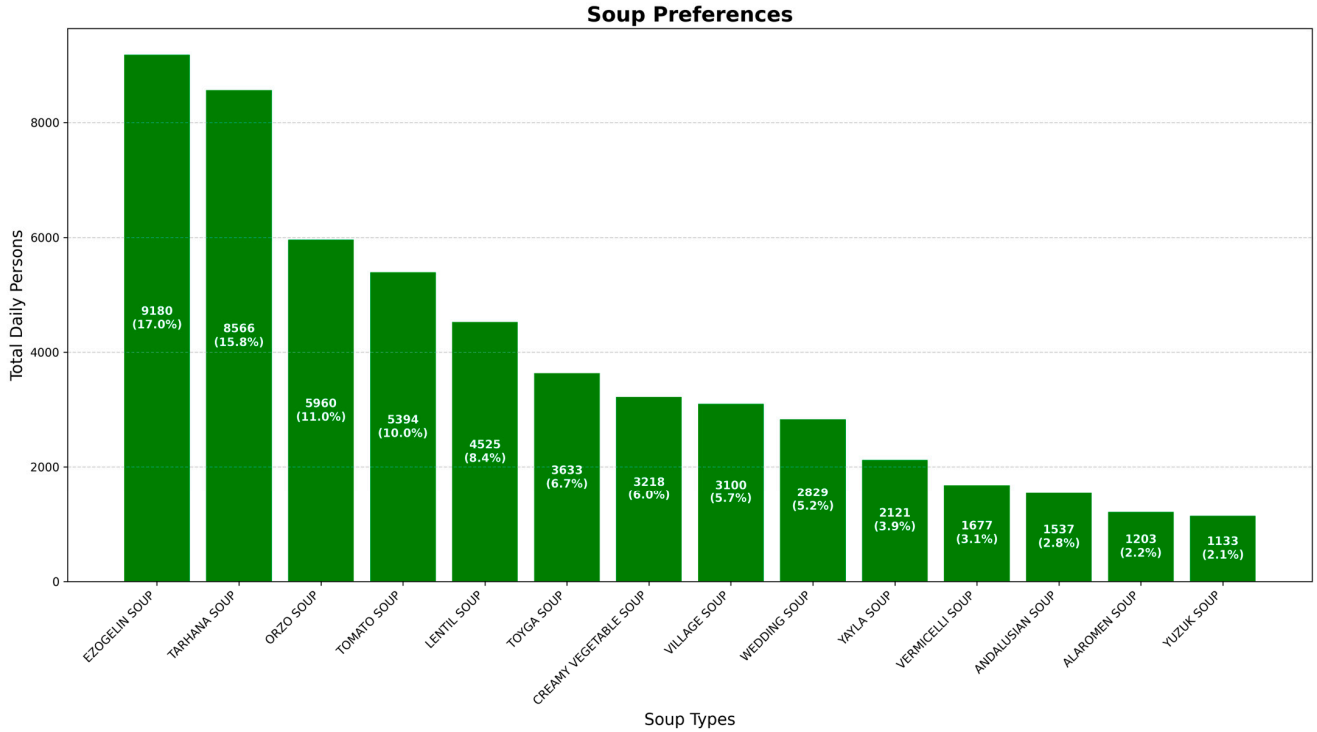


Figure 6. Graph of preference rates for soup.

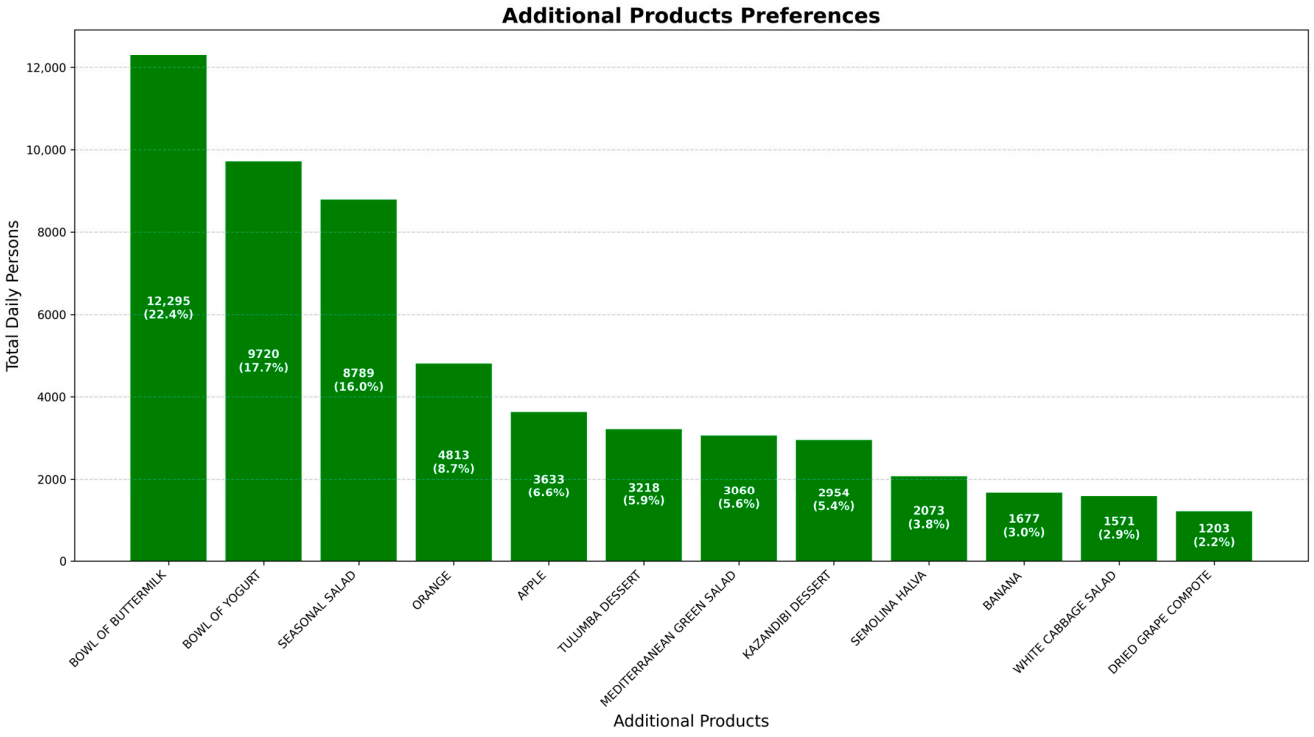


Figure 7. Graph of preference rates for additional products.

Figure 9 presents the correlation graph showing the relationships between the parameters in the original dataset. A series of bar and dot plots visualizing the interactions of daily campus arrival and departure data, average temperature, cafeteria revenue, and the number of people eating in the dining hall reveal row and column cross-interactions. Each of the analyses performed on the dataset has strengthened the dataset and provided a better

understanding of the different dynamics on the Suleyman Demirel University campus, supporting future planning.

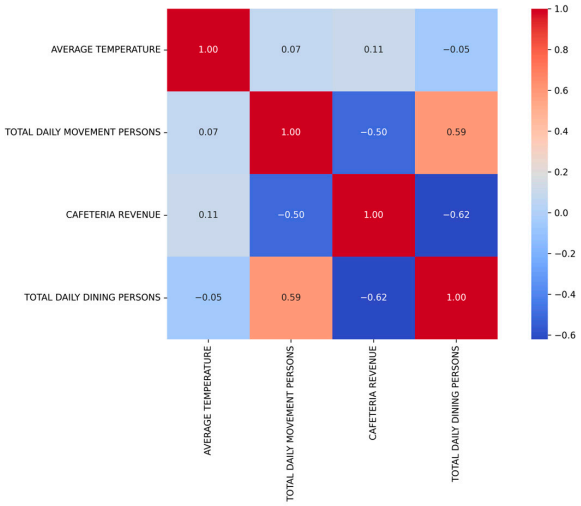


Figure 8. Campus dataset correlation heatmap.

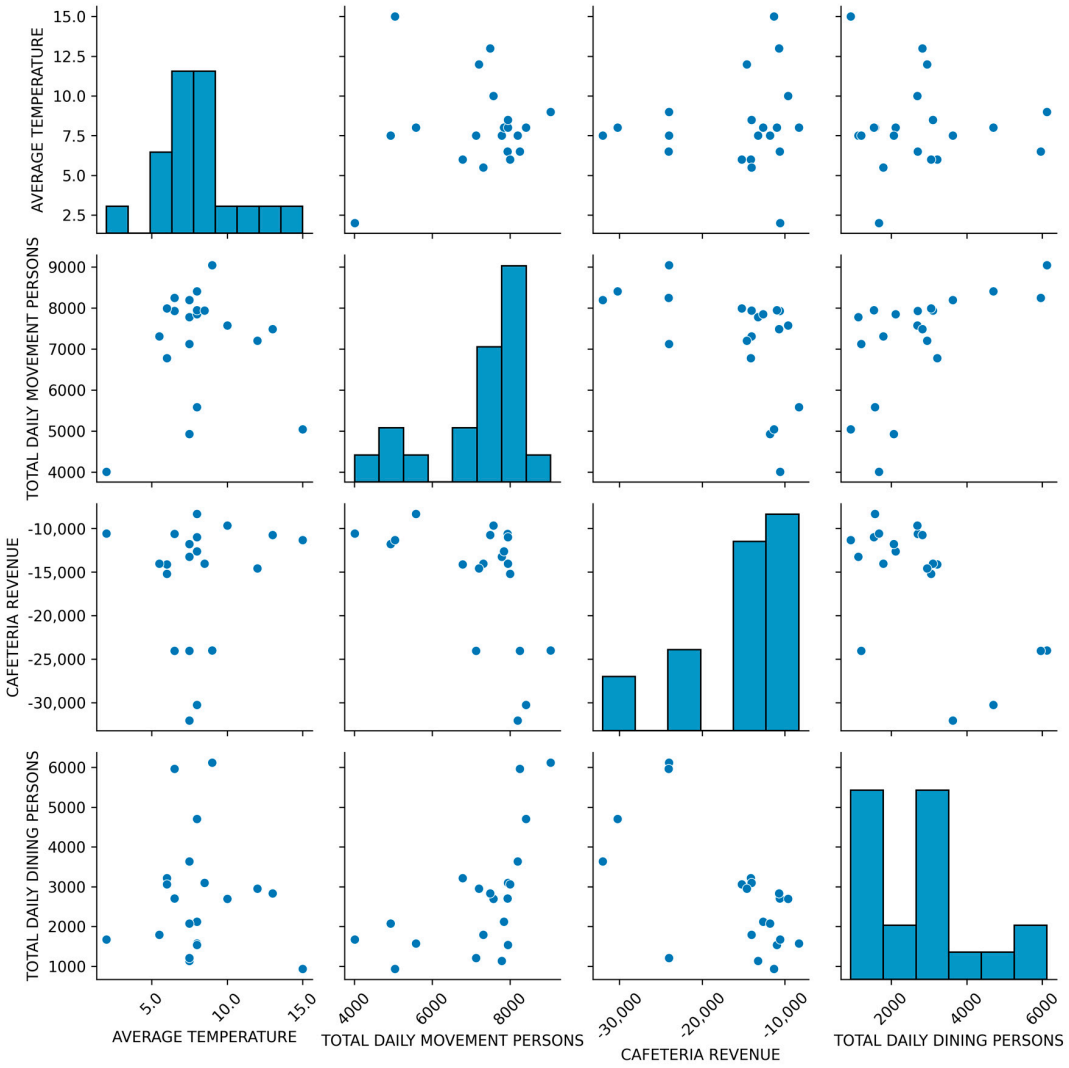


Figure 9. Campus dataset correlation matrix.

3.2. Predictive Performance via Regression Analysis: Machine Learning Results

Changing food demand on campuses due to student and staff density necessitates effective meal planning. The variety of meals served in dining halls has a direct impact on this demand and is important for balancing and optimizing demand. Adjusting the number of meals according to the campus density is essential for preventing waste and efficient use of resources. Preventing food waste not only reduces costs but also contributes to environmental sustainability. Therefore, it is of strategic importance to accurately forecast demand and develop early solutions.

Models were developed using various machine learning regression algorithms to predict the number of meals needed per day in the campus cafeteria. The dataset was expanded with analysis and preprocessing to predict the number of people who will eat on a given day. The dataset includes meal types such as main course one, main course two, soup, and additional products, as well as variables such as air temperature, the number of staff and students eating, the number of people entering the university, other demographic information that is important in determining the density on campus, and canteen turnover. This rich dataset was a valuable resource for forecasting food demand. In the first stage, missing data completion, outlier identification, and data cleaning were performed on the collected data. Then, prediction models were created using different regression algorithms such as Linear Regression, Extra Tree Regressor, Lasso, Decision Tree Regressor, XGB Regressor, and Gradient Boosting Regressor. To evaluate the performance of the models, the dataset was split into two parts, 80% training and 20% testing, to test the prediction accuracy and generalization ability of the model. The performances of these models were evaluated by cross-validation methods, and the model with the highest prediction accuracy was identified. As a result, this study aimed to improve operational efficiency by more accurately forecasting the daily food demand on campus.

Table 2 presents the R^2, RMSE, and MAE values for the regression algorithms compared. The Lasso algorithm achieved the highest accuracy with an R^2 value of 0.999. This success is attributed to its regularization capability, which prevents overfitting by penalizing less significant features. Its low RMSE (1.891) and MAE (1.228) values further emphasize its superior predictive performance. In contrast, Linear Regression demonstrated poor performance due to its inability to handle multicollinearity and irrelevant features. Decision Tree and Extra Tree algorithms showed moderate performance, likely due to their sensitivity to noise and overfitting in smaller datasets. Ensemble methods such as XGBoost performed better than Linear Regression but fell short of Lasso. Gradient Boosting Regressor struggled to effectively capture data relationships, as reflected in its low R^2 value. In summary, Lasso's ability to focus on the most important features underpins its superior accuracy, while the limitations of other models in addressing multicollinearity and feature selection contributed to their lower performance.

The primary factor enabling Lasso to outperform other models is its feature selection and regularization mechanism. This approach reduces the complexity of the model, preventing overfitting while ensuring that only the most relevant variables are included. For instance, features such as "average temperature" exhibited strong coefficients, playing a decisive role in predicting the target variable, "total daily persons." Lasso's simplicity and compatibility with medium-sized datasets further distinguish it from the other models.

Table 2. Regression algorithm results.

Algorithms	R_SQUARED	RMSE	MAE
Lasso	0.999	1.891	1.228
XGBoost Regressor	0.882	547.236	369.643
Decision Tree Regressor	0.765	773.850	592.5
Extra Tree Regressor	0.672	914.830	788.25
Gradient Boosting Regressor	0.549	1072.428	765.745
Linear Regression	0.363	1275.293	910.641

In comparison, XGBoost Regressor achieved a high R^2 value of 0.882 but lagged behind Lasso due to higher RMSE and MAE values. Despite its ability to model complex relationships, XGBoost did not demonstrate superiority in this dataset. Decision Tree and Extra Tree Regressors displayed limited generalization capabilities, as evidenced by their high error rates. Gradient Boosting Regressor and Linear Regression, with their low R^2 values, failed to effectively model the relationships within the dataset. This analysis highlights the effectiveness of Lasso while suggesting that algorithms like XGBoost could achieve better results with optimized hyperparameters. Furthermore, it emphasizes that simpler models may not be suitable when the dataset exhibits complexity, underscoring the importance of aligning model choice with dataset characteristics.

Accurate forecasting of food consumption or actual demand in university dining halls significantly impacts controlling food waste and optimizing resource allocation. After training the dataset, the developed models were tested with real data, and their performance was evaluated. Figures 10 and 11 compare the real and predicted data for each model, with Figure 10 illustrating performance based on the R^2 parameter and Figure 11 using the RMSE parameter. These graphical results align with the accuracy metrics presented in Table 2.

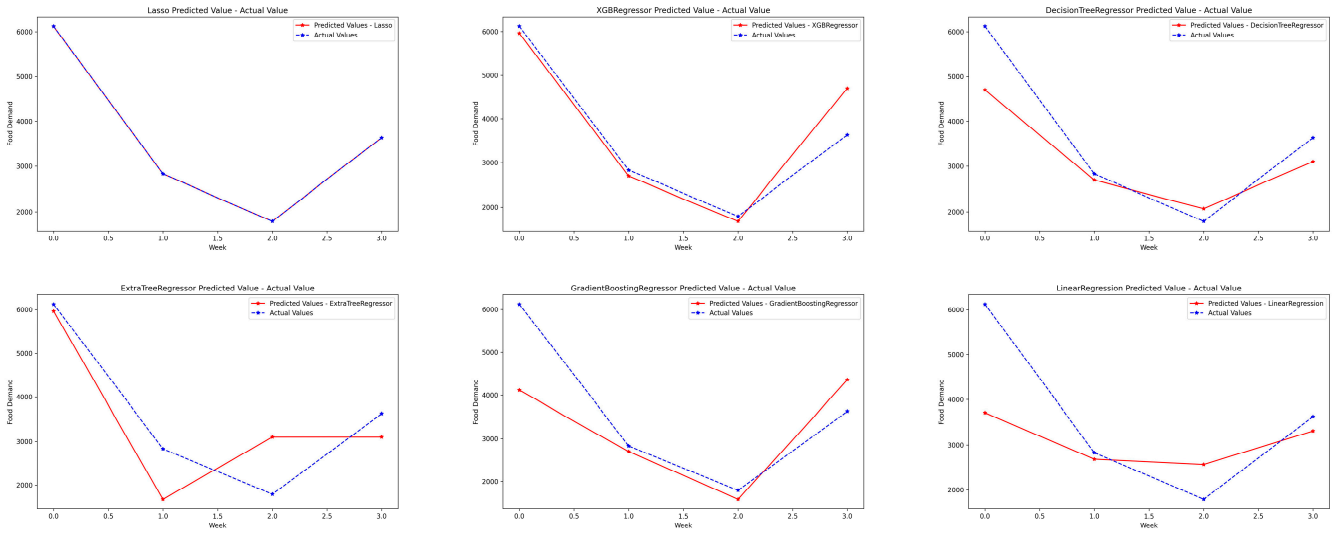


Figure 10. Predicted and actual values of the R^2 parameter.

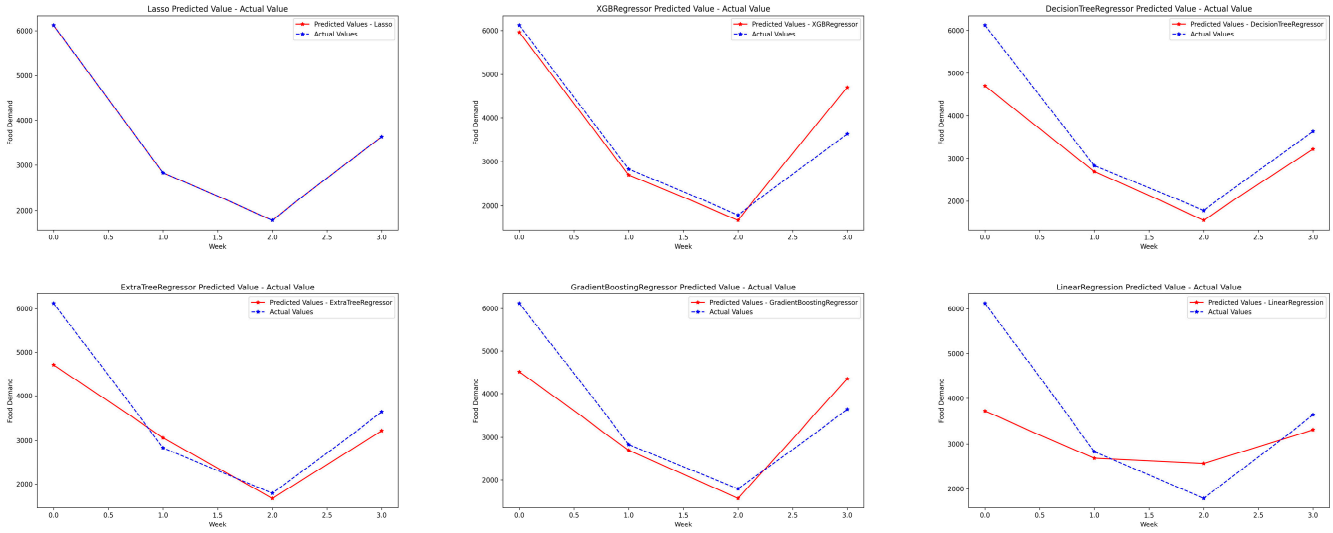


Figure 11. Predicted and actual values of the RMSE parameter.

Figure 10 highlights the predictive accuracy of the models across different data points. The Lasso model demonstrates a near-perfect match between actual and predicted values, evident from its consistently high R^2 values across data points. In contrast, the Linear Regression model shows significant deviations, reflecting its poor ability to handle data variability and multicollinearity. The Decision Tree Regressor and Extra Tree Regressor models exhibit moderate performance, with occasional fluctuations in accuracy, indicating sensitivity to noise and overfitting.

Figure 11 focuses on RMSE values, measuring the average magnitude of error in the predictions. The Lasso model achieves the lowest RMSE values across the dataset, confirming its ability to make precise predictions. On the other hand, models like Gradient Boosting Regressor and Linear Regression display higher RMSE values, suggesting difficulties in capturing complex patterns. These results validate the Lasso model's superiority in demand forecasting and highlight the limitations of other models in handling noisy or complex data, emphasizing the need for careful algorithm selection to minimize food waste and improve sustainability.

In this study, the dataset was divided into training and test sets with a ratio of 80% and 20%, respectively. Cross-validation is utilized as a key technique to evaluate the overall performance of the models. This method is applied to assess how well each model generalizes the data without encountering issues like overfitting or underfitting. By splitting the dataset into different subsets, cross-validation enables testing the models' accuracy across these partitions and provides reliable insights into their effectiveness.

Figure 12 depicts the variation in machine learning model performance across different cross-validation fold counts, ranging from 5 to 10. The Lasso model consistently demonstrated the best performance across all validation folds, standing out with its high generalization capacity and low tendency for overfitting. In contrast, the Linear Regression model exhibited the poorest performance, with limited generalization ability. Decision Tree and Extra Tree models showed fluctuating performance, particularly prone to overfitting on smaller datasets. XGBoost and Gradient Boosting models delivered moderate performance but were sensitive to hyperparameter tuning. This graph highlights the importance of cross-validation as a tool for evaluating the generalization ability of models and emphasizes the need to consider both consistency and accuracy when selecting the best model.

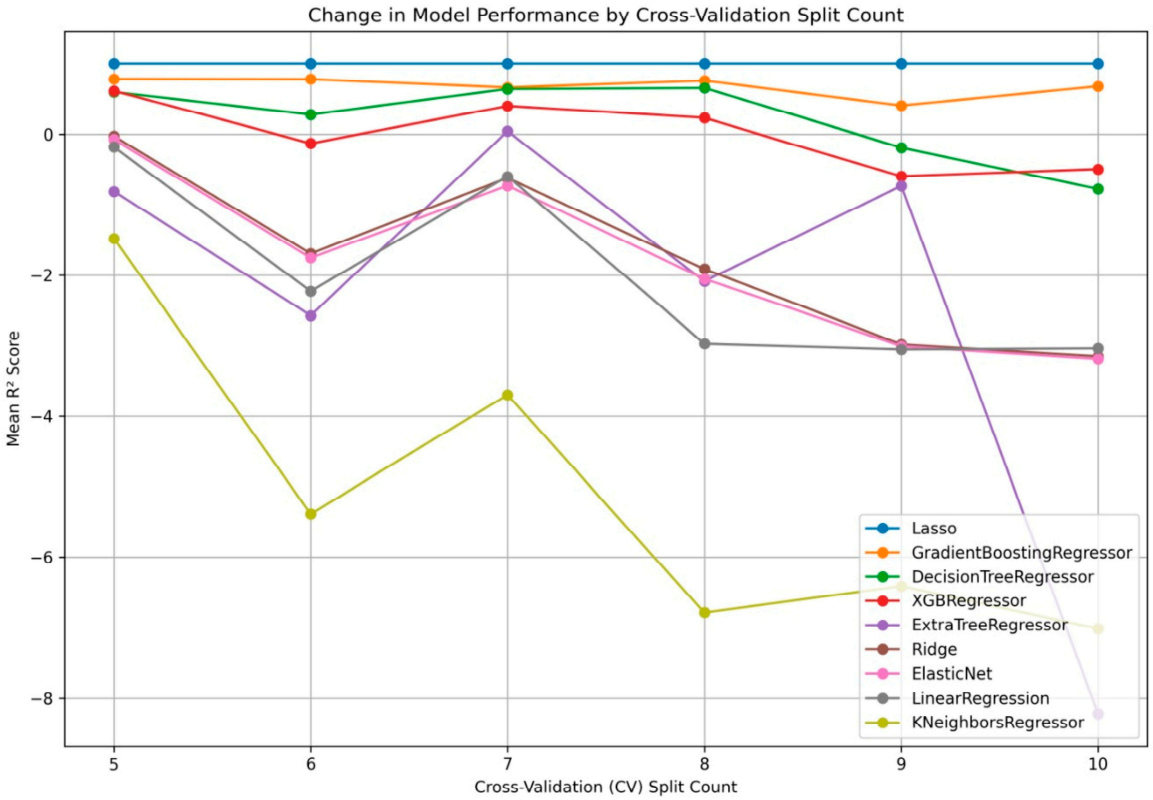


Figure 12. Change in model performance by cross-validation split count.

Figure 13 highlights the changes in the average R^2 scores of the models and demonstrates their consistency in performance. The Lasso model consistently achieves the highest and most stable R^2 scores across all folds, showcasing its strong generalization ability and effectiveness in handling overfitting. This graph emphasizes that Lasso is the most reliable model for this dataset and highlights the need for improvements in the generalization capabilities of other models. Figure 12 emphasizes consistency, while Figure 13 provides a clearer depiction of differences in model performance for specific cross-validation split counts. When used together, they offer a more comprehensive analysis of overall model performance.

Variability in food demand due to student and staff density necessitates effective food planning on campuses. The variety of meals served in dining halls is critical for balancing and optimizing demand, preventing waste and ensuring efficient use of resources. Accurate forecasting of demand ensures that student and staff needs are met, while monitoring catering services contributes to optimizing labor costs and capacity planning. In this study, the menus that have the greatest impact on consumption were identified, and the impact of each dish on demand was analyzed in detail by dividing them into different categories. Demand forecasting models created using various machine learning methods were enriched with menu data, and as a result, the most successful model was identified.

The integration of machine learning models to balance food production and demand has achieved a significant improvement by reducing food waste to 28%. This result indicates that on a campus producing 1000 portions of food daily, only 280 portions are wasted, while the remaining 720 portions are effectively consumed. This reduction not only decreases the carbon footprint from an environmental sustainability perspective but also enhances cost efficiency. The implementation of the model establishes a strategic foundation for resource optimization in dining services by preventing unnecessary production. This highlights that

while waste cannot be entirely eliminated, it can be significantly reduced by improving current operations.

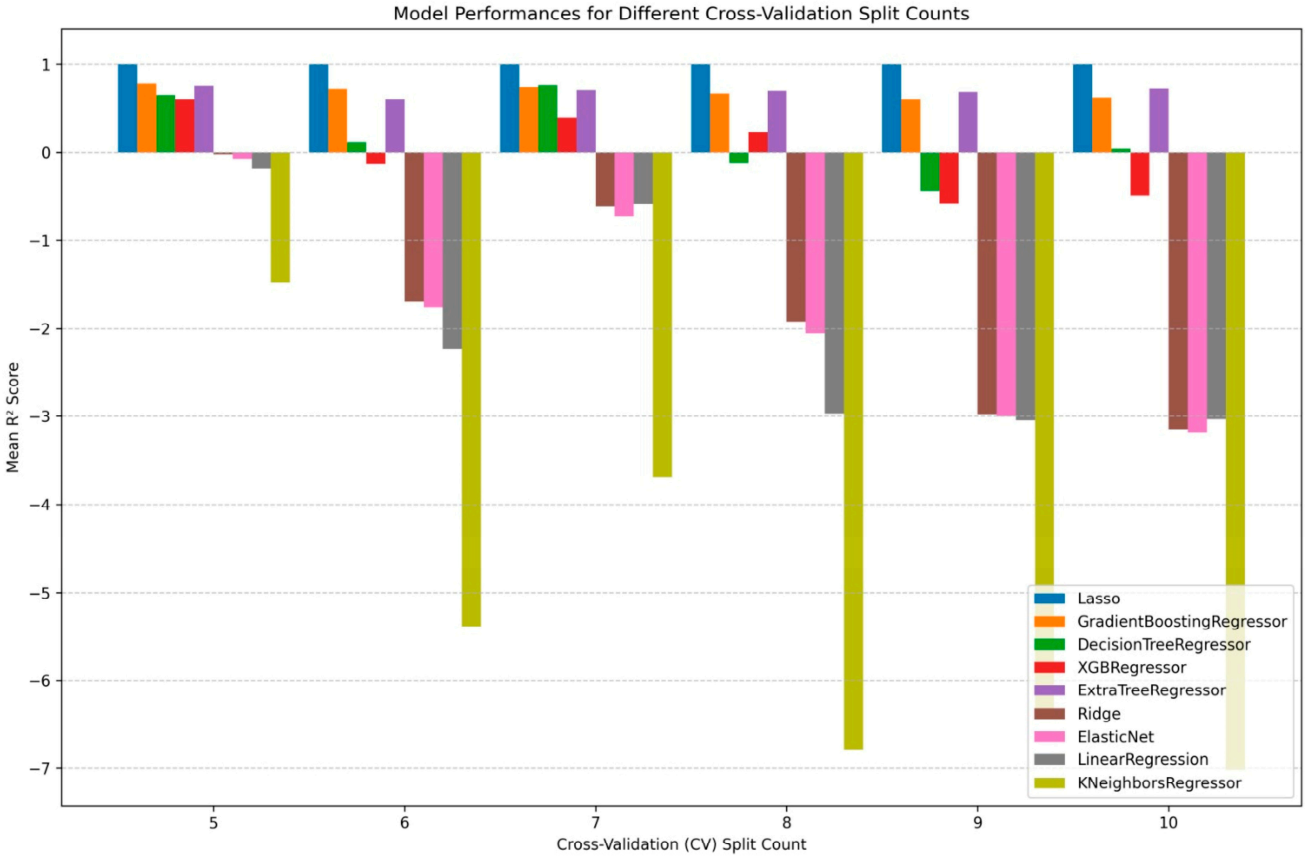


Figure 13. Model performances for different cross-validation split counts.

Campus administrators can effectively utilize existing real-time data (e.g., temperature, daily entry counts, student and staff density) to implement the model proposed in the article. Considering that campus data at Suleyman Demirel University is already regularly collected and accessible through an interface in real-time, integrating temperature and menu information from web services along with daily entry-exit records allows for the use of a demand forecasting model. Through this integration, administrators can accurately predict daily food demand and plan food production accordingly. Additionally, during special occasions such as university events or exam days, they can monitor demand spikes in real time and optimize service capacity. This approach aims to enhance operational efficiency while significantly reducing food waste.

The findings show that not only food count forecasts are successful, but also that the efficiency of dining hall services can be increased if the menus are optimized correctly.

The experimental results are as follows:

1. The training sample is sufficient and the prediction accuracy is significantly improved with the regression algorithms;
2. It is difficult to estimate the true value of the dining hall consumption data, and it is found that the impressiveness of the menus is higher when the total number of consumers is taken into account;
3. It is an innovative idea that consumers have the potential to organize menus in advance by determining the preference rates of food types on campus.

The impact of the experimental results is as follows:

1. Additional staff can be assigned to increase service speed and reduce customer waiting times on peak food consumption days;
2. Additional staff can be included and additional products can be offered on days when cafeteria usage is expected to increase;
3. By determining the preference rates and variety of food options on the menus, campus density can be monitored, and menus can be tailored according to the days of the week.

4. Discussion

In a study, researchers developed a demand forecasting model for a university dining hall, taking into account the calendar effect and turnstile passage data, and examined eighteen Machine Learning models collected under five main techniques. They tested the diversity of methods with three Artificial Neural Network models, four Gaussian Process Regression models, six Support Vector Regression models, three Regression Tree models, two Ensemble Decision Tree (EDT) models, and one Linear Regression model. The EDT Boosted model achieved the best prediction performance (MSE = 0.51, MAE = 0.50, and R = 0.96) [22]. Another study analyzed plate shape and size as a physical factor in reducing waste based on the number of meals predicted by machine learning algorithms and found that the use of oval plates was effective in reducing waste significantly [20]. In addition, predictions were made using food tracking data to ensure that food is preserved during storage. The Random Forest Optimizer method was used to predict the freshness of the food in the storage area, and the best result was obtained with an accuracy of 94.01% [15]. In a study on food waste planning, ANN was used to develop a prediction, taking into account the number of reservations, but not the number of cancellations. This situation created uncertainty between reality and prediction [41].

In the food industry, machine learning-based demand forecasting models have been used to manage the supply chain more effectively. In studies using LSTM and SVM algorithms, the success of LSTM forecasts was determined as a result of the comparison of MAE and MAPE values [42–45]. In this study, regression models are preferred because time series models require a lot of data and long training time. Regression models offer simplicity, speed, and interpretability, enabling fast predictions with small data sets. Their low computational cost and clear outputs benefit decision-makers.

Machine learning regression algorithms have been used in food supply chain processes [17,30]. In this study, in addition to the features in the literature, different features were added to the dataset created for on-campus density analysis and food demand forecasting by tracking food consumption operations. While it was predicted that the number of people entering the campus and the number of foods consumed would be linear, with the features added to the dataset, it was determined that the biggest factor in the number of foods actually consumed is the foods included in the menu. The effect of the percentages of the preferred foods on the results was seen by dividing the foods into 4 categories in the menus. In addition, the effect of the data of the cafeteria selling snack foods on the campus was also evaluated. The results of the successful model developed with ML regression algorithms based on the campus dataset were evaluated with RMSE, MAE, and R^2 parameters in predicting the number of food demands, and the R^2 coefficient ranged from 0.99 (Lasso) to 0.36 (Linear Regression). It was seen that the Lasso algorithm created the most successful model due to its ability to perform effective feature selection and regularization, reducing model complexity and preventing overfitting, which ensures the inclusion of only the most relevant variables in predictions.

Compared to the studies in the literature, they have addressed food waste in universities and used data such as reservation, plate shape, behavioral impact, and turnstile entry

only, but there has not been a comprehensive study on demand forecasting based on the number of meals consumed in the cafeteria, meal types, and food types by tracking on-campus mobility. There is a research gap in the context of on-campus process management where meal counts are influential. Therefore, one of the main contributions of the current study is to fill this gap by creating a comprehensive dataset that includes campus density, temperature, other canteen data, and menus that affect meal counts, and detailed analysis for university management.

Moreover, recent studies underline the importance of sustainability and food waste reduction. Zhang and Kwon (2022) demonstrated that implementing trayless dining significantly reduces food waste while positively affecting diner satisfaction and food selection behaviour [21]. Additionally, Martin-Rios et al. (2020) highlighted the role of sustainability-oriented innovations in food waste management technology, aligning with circular economy principles [11]. Building on these insights, the proposed model contributes by effectively balancing production and demand, reducing food waste by up to 28%. This approach minimizes unnecessary production, optimizes resource allocation, and supports sustainable practices in campus dining services.

In addition to the aspects examined in previous studies for demand forecasting, in the proposed model, in addition to meal counts, other canteen consumption data on campus and preference rates of menus that affect demand are also taken into account. The current research used current on-campus data to estimate the volume of available meals. The novelty of this study is that it analyzes on-campus density, daily temperature, canteen data, days of the week, and daily dining hall attendance rates, and provides a high-performing model with the segmentation of menus that is unprecedented in previous research. During periods of increased demand, such as exam days or university events, the proposed model leverages real-time input data and historical demand trends to more accurately analyze demand dynamics. This allows managers to anticipate demand surges, include popular meal options in the menu, and effectively optimize production planning. The unique contribution of this study lies in the integration of menu-related variables into predictive models, which significantly contributes to operational processes such as reducing food waste, improving resource management, enhancing student satisfaction, and optimizing menu planning.

The proposed model has managed to create a remarkable potential to improve the overall service quality and efficiency within the scope of managerial operations of the mentioned university, such as food waste reduction, resource management, student satisfaction, menu planning, canteen improvement. The proposed solution is not limited to a meal count prediction model but is capable of designing and solving different objectives with different input data in any campus area.

Limitations and Further Research

The demand forecasting model developed in this study holds significant potential to improve operational efficiency and reduce food waste in campus dining services. However, limitations such as the lack of long-term data and cultural differences between campuses constrain the model's generalizability. Future research could address these limitations by testing the model in different universities and regional contexts. Additionally, long-term pilot implementations could provide more comprehensive evaluations of the model's sustainability.

Integrating real-time data can enhance prediction accuracy and enable menu optimization based on daily demand. At Suleyman Demirel University, the existing data system already facilitates access to temperature, menu, and entry-exit records through an interface. This integration allows administrators to accurately predict demand and optimize service

capacity during special occasions such as events or exam days. Consequently, operational efficiency can be improved while significantly reducing food waste. The model is not only a tool for predicting meal demand but also a strategic asset for resource management, menu planning, and improving student satisfaction.

Future research could enhance the versatility and real-world applicability of the proposed demand forecasting model by integrating factors such as cultural preferences, multi-campus comparisons, and cost-related impacts. Investigating the role of religious and national holidays, alongside variations in meal preferences across regions, could offer valuable insights. By incorporating these elements, the model could better account for regional and cultural dynamics in demand prediction and menu customization. Furthermore, analyzing the model's performance across multiple campuses could highlight its robustness and adaptability, while addressing cost considerations would enhance its practicality in resource distribution and operational planning.

5. Conclusions

Effective planning and management of campus food services are critical to reducing food waste and improving service quality. This study utilized machine learning algorithms to forecast food demand based on campus density, dining hall entries, and menu preferences. The Lasso algorithm outperformed others with an R^2 value of 0.99, enabling accurate demand predictions and better alignment of production with consumption needs. The unique contribution of this study lies in the integration of menu-related variables into predictive models. Results showed that food consumption is influenced more by menu design than campus density, emphasizing the importance of optimizing menu offerings to meet preferences. This study incorporates not only meal counts but also other canteen consumption data and preference rates of menus, enabling a more accurate modeling of demand. During periods of increased demand, such as exam days or university events, the model effectively used real-time input data and historical demand trends to better analyze demand dynamics. In addition to enhancing service quality, the forecasting models significantly minimized food waste by preventing overproduction and reducing resource wastage. By balancing production with demand, this model significantly reduced food waste to 28%. The study highlights its contribution to reducing food waste by simultaneously lowering operational costs and supporting environmental sustainability goals. Dynamic and responsive predictions enabled by real-time data integration made campus food services more adaptable to changing demands. Future research should enhance the generalizability of the proposed demand forecasting model by incorporating variables such as cultural preferences, multi-campus analyses, and cost implications. Examining the impact of religious and national holidays, regional meal preferences, and menu segmentation could provide valuable insights. Expanding the model to multi-campus systems and exploring cost-related variables would further improve its adaptability and effectiveness. This study underscores the value of data-driven approaches in achieving sustainable and efficient food service operations, providing a solid foundation for innovative strategies in higher education.

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